

SUMS OF RANDOM VARIABLES

$$\text{SUMS} \quad X_1 + X_2 + \dots + X_N = S_N$$

$$\text{MEANS} \quad \frac{1}{N} (X_1 + X_2 + \dots + X_N) = M_N$$

$$E(X_1 + \dots + X_N) = \sum_{i=1}^N E(X_i)$$

$$E\left(\frac{1}{N} (X_1 + \dots + X_N)\right) = \frac{1}{N} \sum_{i=1}^N E(X_i)$$

$$\text{Var}(X_1 + X_2 + \dots + X_N)$$

"

$$E((X_1 + X_2 + \dots + X_N)^2) -$$

"

$$\begin{aligned} & \underline{E(X_1^2)} + \underline{X_1 X_2} + \dots + X_1 X_N + \\ & X_1 X_2 + \underline{X_2^2} + \dots + X_2 X_N \\ & \vdots \end{aligned}$$

N^2 TERMS

$$X_N^2))$$

$$= \sum_{i=1}^N \sum_{j=1}^N E(X_i X_j) - E(X_i)E(X_j)$$

$$\begin{aligned} \text{Var}(Y) &= \\ E(Y^2) - (E(Y))^2 \end{aligned}$$

$$[E(X_1 + \dots + X_N)]^2$$



$$\begin{aligned} & (E(X_1) + \dots + E(X_N))^2 \\ & \underline{(E(X_1))^2} + \underline{E(X_1)E(X_2)} + \dots \end{aligned}$$

N^2 TERMS

$$= \sum_{i=1}^N \sum_{j=1}^N \text{Cov}(X_i, X_j)$$

$$= \sum_{i=1}^N \text{Var}(X_i) + \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \text{Cov}(X_i, X_j)$$

THIS IS THE GENERAL CASE, BUT WE
 MOST OFTEN CARE ABOUT WHEN
 THE $\{X_i\}$ ARE I.I.D. : INDEPENDENTLY
 AND IDENTICALLY DISTRIBUTED.

(ALL HAVE SAME PDF AND ARE INDEPENDENT.)

WHEN $X_1, \dots, X_N$ ARE I.I.D,

$$E(X_i) = \mu, \quad \text{Var}(X_i) = \sigma^2 \quad \text{FOR ALL } i.$$

$$E\left(\sum_{i=1}^N X_i\right) = N\mu$$

$$E\left(\frac{1}{N} \sum_{i=1}^N X_i\right) = \mu$$

$$\text{Var}\left(\sum_{i=1}^N X_i\right) = N\sigma^2$$

$$\text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) = \frac{1}{N^2} (N\sigma^2) = \frac{1}{N} \sigma^2$$

$$\lim_{N \rightarrow \infty} \text{Var}(\text{MEAN}) = 0$$

← (*)
IMPORTANT
FOR
REAL
WORLD.

(*) THE MEAN OF A BUNCH OF
NOISY OBSERVATIONS GETS LESS
AND LESS NOISY THE MORE WE HAVE.

HOW FAST DO WE GET THERE?

CHEBYSHEV'S INEQUALITY FROM EARLIER:

$$P(|M_n - \mu| \geq \epsilon) \leq \frac{1}{n\epsilon^2} \sigma^2$$

SAMPLE MEAN IS MORE
THAN ϵ AWAY FROM
TRUE MEAN

GETS SMALLER
AS n GROWS.

EX $X =$ NUMBER OF TORTOISES ON A
GIVEN WALK. ~~Poisson~~
POISSON WITH $\alpha = 5$.

$P(\text{SEE BETWEEN 40 AND 60 TORTOISES}$
 $\text{ON 10 INDEPENDENT WALKS})$

$$S_{10} = \sum_{i=1}^{10} X_i$$

$$E(S_{10}) = 10 \cdot 5 = 50$$

$$P(|S_{10} - 50| \leq 10) = ?$$

$$P(|M_{10} - 5| \leq \boxed{1}) =$$

$$Pr. \geq 1 - \frac{1}{n \epsilon^2} \sigma^2$$

$\uparrow$ $\uparrow$
 10 1^2

VARIANCE OF X_i
 For Poisson, $\sigma^2 = \alpha = 5$

$$\boxed{\geq} 1 - \frac{5}{10} = \frac{1}{2}$$

THE EXACT ANSWER WOULD REQUIRE

THE JOINT PDF OF $(X_1, X_2, \dots, X_{10})$

↳ THE PDF OF $X_1 + X_2 + \dots + X_{10}$

WHAT IS THAT?

WHAT'S THE PDF OF $X_1 + X_2$
(GENERAL CASE)?

$F_Z(z)$ $Z = X + Y$ CDF of $Z \Rightarrow \frac{d}{dz}$ PDF of Z .

" CDF

$$P(Z \leq z) = P(X + Y \leq z)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x, y) dy dx$$

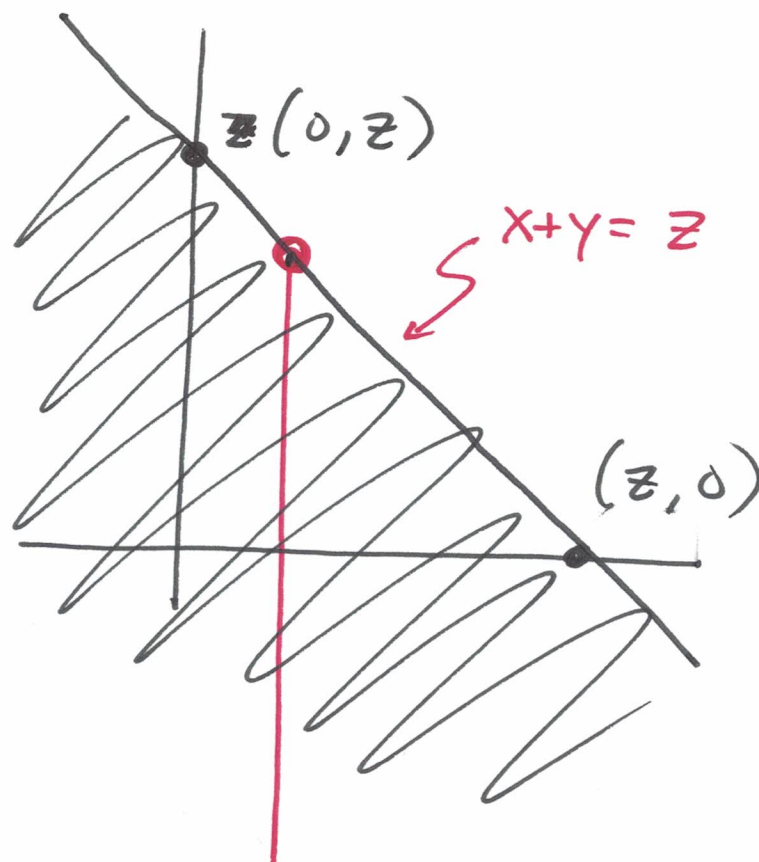
(A red arrow points from the circled $z-x$ to the upper limit of the inner integral.)



PDF

$$\frac{d}{dz} = \int_{-\infty}^{\infty} f_{XY}(x, z-x) dx$$

$f_Z(z)$



$$f_z(z) = \int_{-\infty}^{\infty} f_{xy}(x, z-x) dx$$

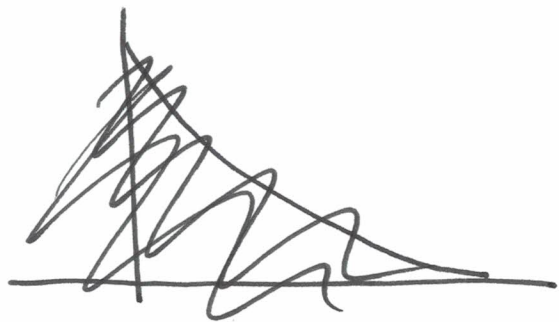
$\hookrightarrow f_{xy} = f_x f_y$

IF X AND Y ARE INDEPENDENT,

$$f_z(z) = \int_{-\infty}^{\infty} \underbrace{f_x(x)}_{\text{PDF OF } x} \underbrace{f_y(\textcircled{z}-x)}_{\text{PDF OF } y} dx$$

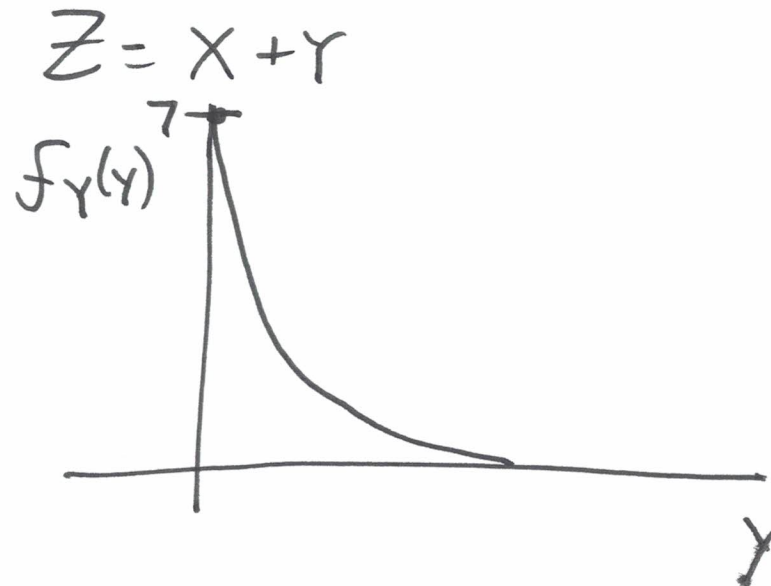
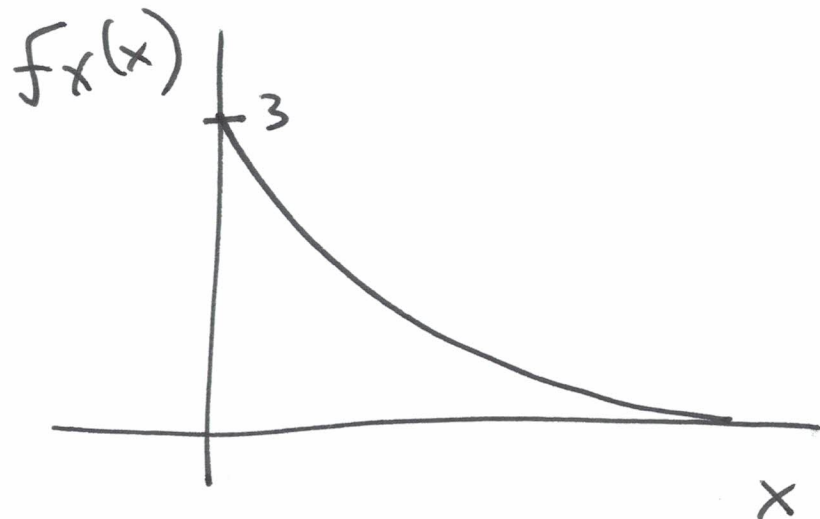
THE CONVOLUTION OF f_x AND f_y !
 (RELATED TO SIGNALS AND SYSTEMS)

WHAT DOES THIS INTEGRAL ACTUALLY MEAN?
HOW TO COMPUTE IT?



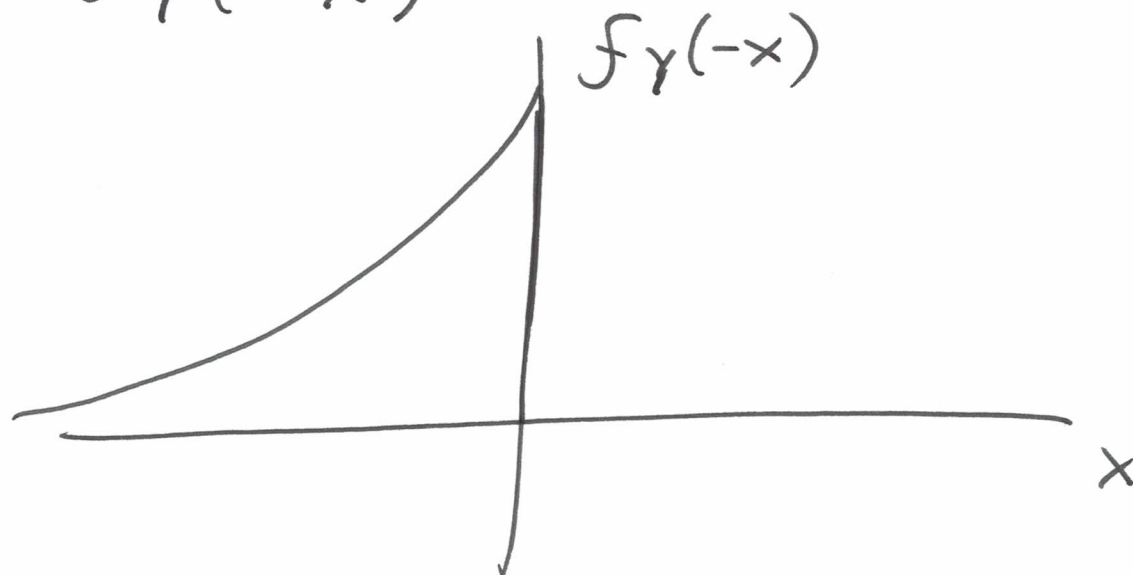
EX X IS EXPONENTIAL
WITH $\lambda_x = 3$
 Y IS EXPONENTIAL
WITH $\lambda_y = 7$

INDEP.

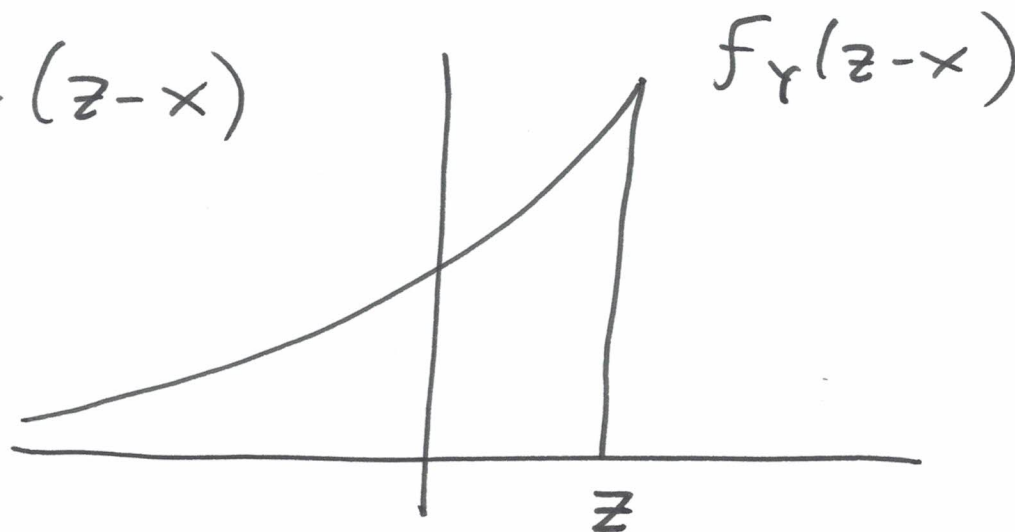


WHAT IS $f_Y(z-x)$?

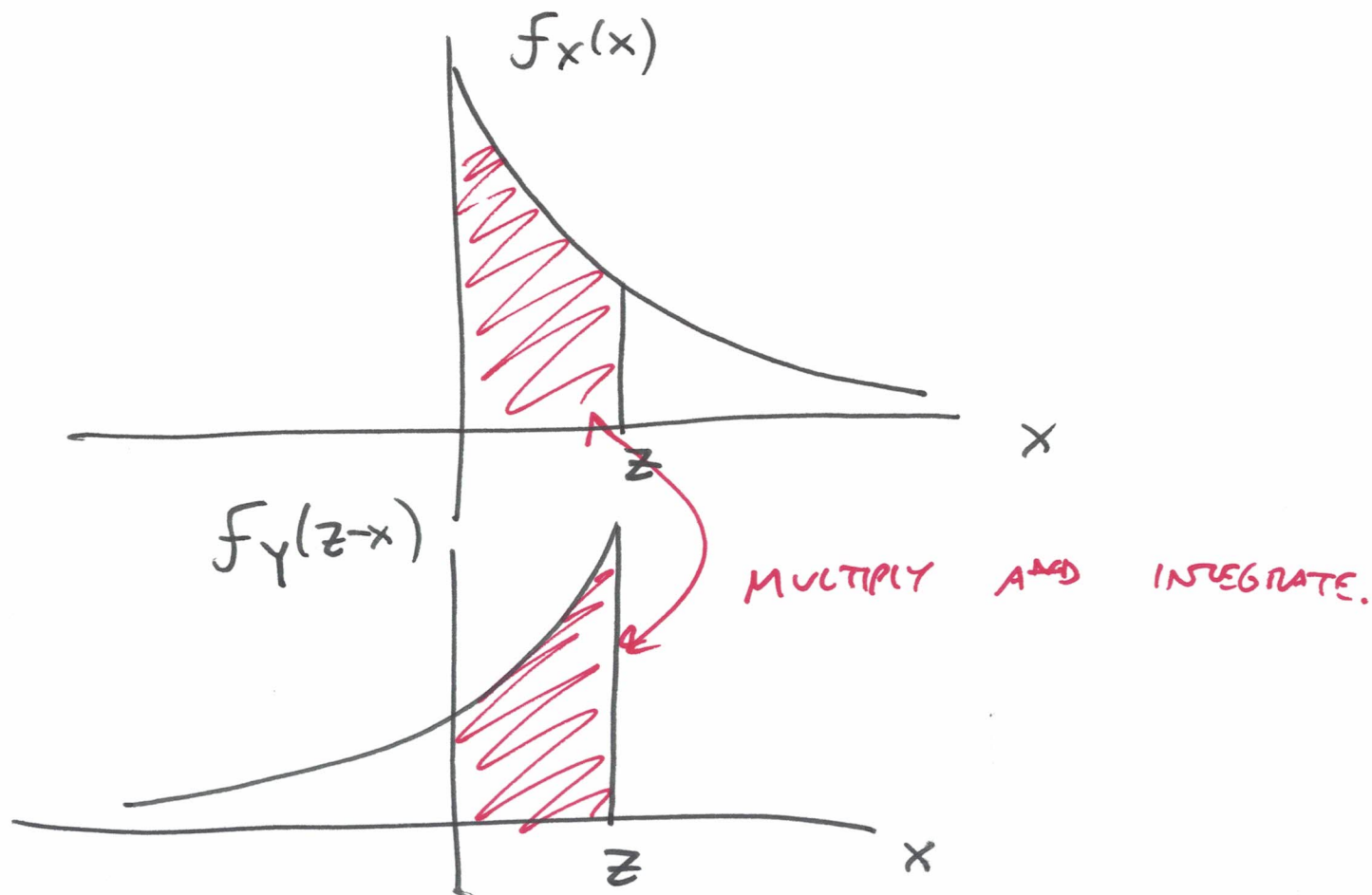
FIRST, $f_Y(-x)$



THEN $f_Y(z-x)$



$$\text{So } \int f_X(x) f_Y(z-x)$$



$$\int_{-\infty}^{\infty} f_x(x) f_y(z-x)$$

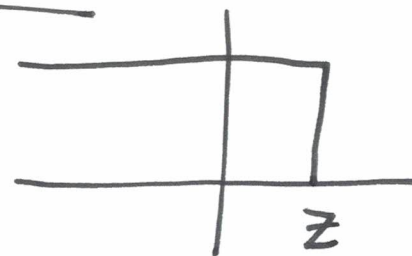
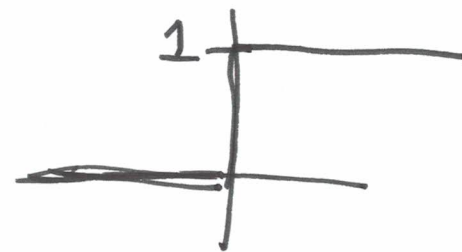


$$3e^{-3x} \underline{u(x)} 7e^{-7(z-x)} \underline{u(z-x)} dx$$

$$\int_0^z 3e^{-3x} \cdot 7e^{-7(z-x)}$$

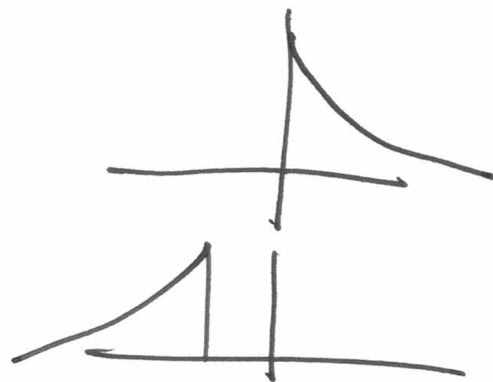
STEP FUNCTION

$u(x)$



WHEN $z < 0$, NO OVERLAP.

WHEN $z \geq 0$, OVERLAP



$$f_z(z) = \int_0^z 3e^{-3x} 7e^{-7(z-x)} dx$$

$$= 21e^{-7z} \int_0^z e^{+4x} dx$$

$$= \frac{21}{4} e^{-7z} \left(e^{4x} \right)_{x=0}^{x=z}$$

$$= \frac{21}{4} e^{-3z} - \frac{21}{4} e^{-7z} \quad z \geq 0.$$

IN GENERAL, THE PDF OF INDEPENDENT

$$X_1 + X_2 + \dots + X_N$$

would BE $f_{X_1} * f_{X_2} * f_{X_3} \dots * f_{X_N}$

$\uparrow$
CONVOLUTION

YUCK!

THIS IS A LOT EASIER IN

THE FREQUENCY DOMAIN USING

FOURIER TRANSFORMS

(DON'T WORRY
ABOUT THIS PAGE!)

THIS SEEMS LIKE A PAIN.

ISNT THERE AN EASIER WAY

TO TALK ABOUT THE PDF

OF THE SUM OR THE MEAN?

YES! NEXT TIME: CENTRAL
LIMIT THEOREM

PREVIEW $f_{X_1} * f_{X_2} * \dots * f_{X_N} \longrightarrow \text{GAUSSIAN} (?!)$

COMING BACK TO THE FIRST PART:

THE WEAK LAW OF LARGE NUMBERS.

$$\lim_{n \rightarrow \infty} P(|M_n - \mu| < \varepsilon) = 1$$

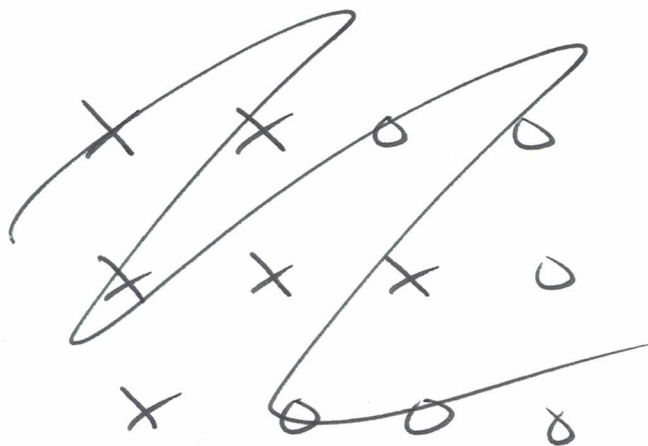
↑
SAMPLE
MEAN↑
TRUE
MEAN↑
ANY SMALL
TOLERANCE

OUR SAMPLE MEAN WILL ALWAYS BECOME
ARBITRARILY CLOSE TO ~~TO~~ THE TRUE MEAN.

LIKE "THE LAW OF AVERAGES";
EVERYTHING EVENTUALLY MUST CONVERGE TO
THE TRUTH.

STRONG LAW OF LARGE NUMBERS:

$$P\left(\lim_{n \rightarrow \infty} M_n = \mu\right) = \underline{1}.$$



THINGS LIKE CHERBYSEV

GIVE RATES FOR HOW FAST THIS HAPPENS.